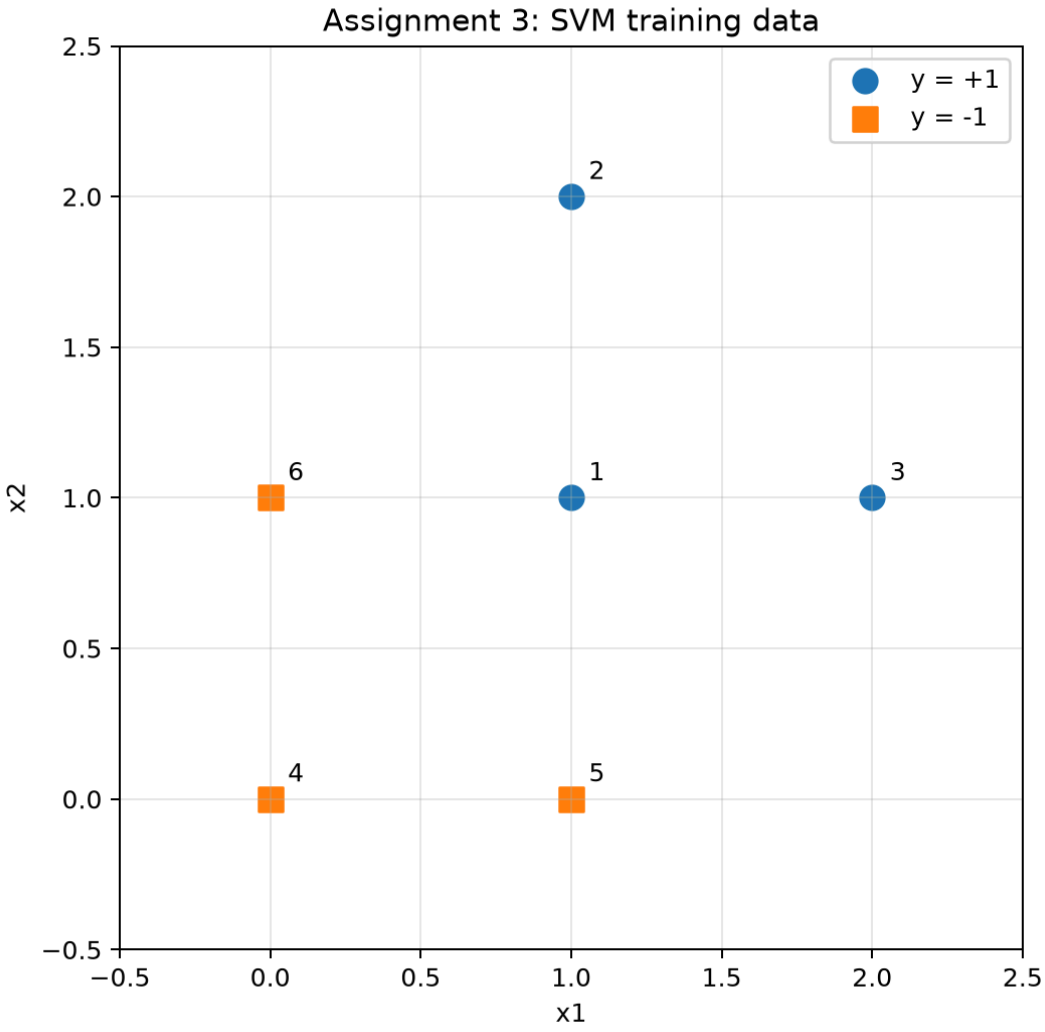
1A)



Looking at it by eye, it appears that pts 1,5,6 are the support vectors since they lie closest to the opposite label and thus the decision boundary.

1B)

HW 3

1b) $X = (1,1)^1, (1,2)^2, (2,1)^3, (0,0)^4, (1,0)^5, (0,1)^6$

$y = +, +, +, -, -, -$

Guess: support vectors are 1, 5, 6, so $\alpha_2 = \alpha_3 = \alpha_4 = 0$

$y_i(w \cdot x_i + b) = 1$

$w = \sum_i \alpha_i y_i x_i$ & $\sum_i \alpha_i y_i = 0$

$(2,2) = \alpha_1(1,1) - \alpha_5(1,0) - \alpha_6(0,1)$

p1: $w_1 + w_2 + b = 1$ $\quad 0 = \alpha_1 - \alpha_5 - \alpha_6$

p5: $w_1 + b = -1$ $\quad 2 = \alpha_1 - \alpha_5$

p6: $w_2 + b = -1$ $\quad 2 = \alpha_1 - \alpha_6$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{array}\right]$$

$w = (2,2)$

$b = -3$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 1 & -1 & 0 & 2 \\ 1 & 0 & -1 & 2 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\alpha_1 = 4$ $\quad \alpha_2 = 0$

$\alpha_5 = 2$ $\quad \alpha_3 = 0$

$\alpha_6 = 2$ $\quad \alpha_4 = 0$

$4 - 2 - 2 = 0$

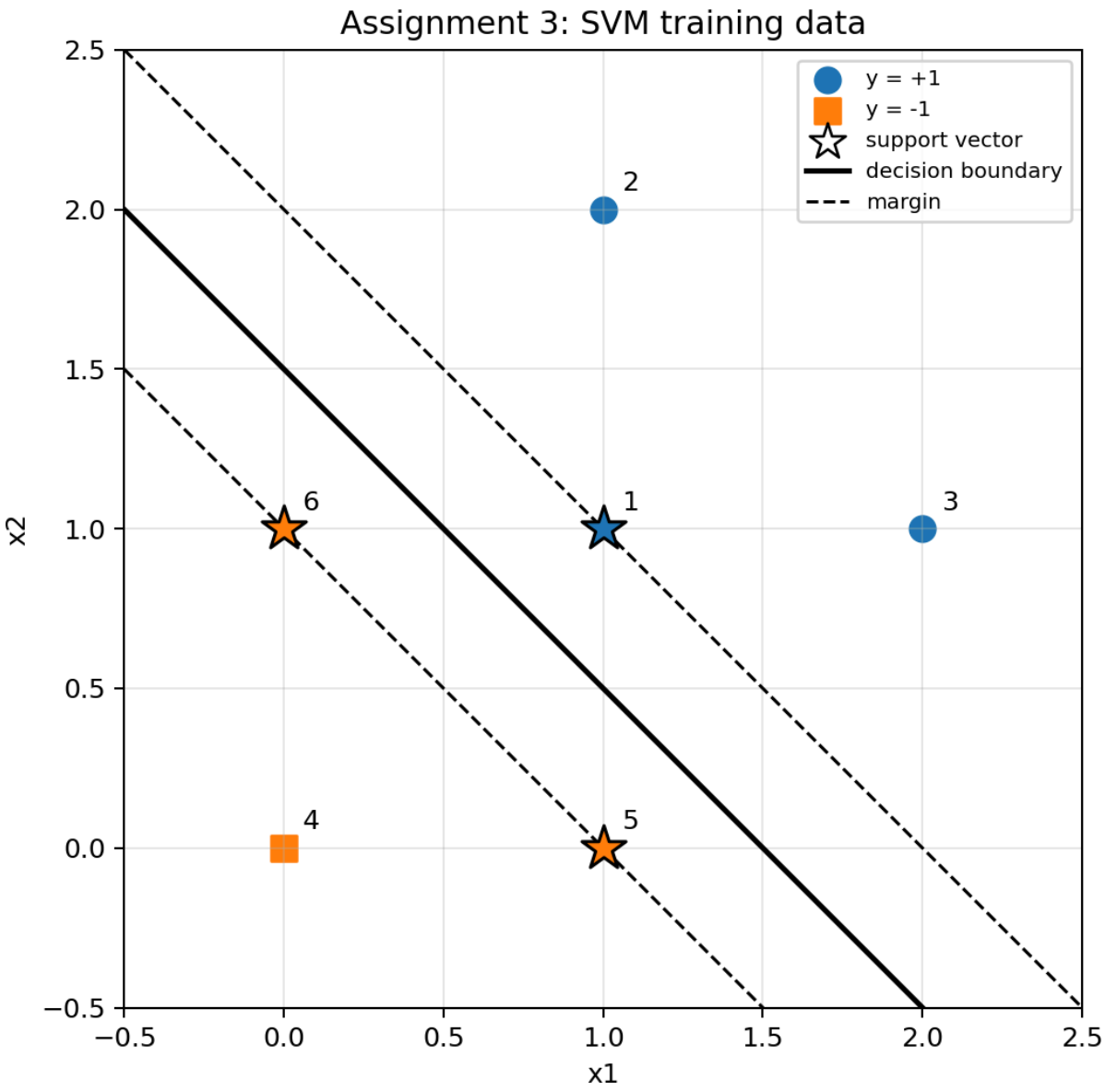
$\|w\| = 2\sqrt{2}$

margin width $= \frac{\sqrt{2}}{2}$

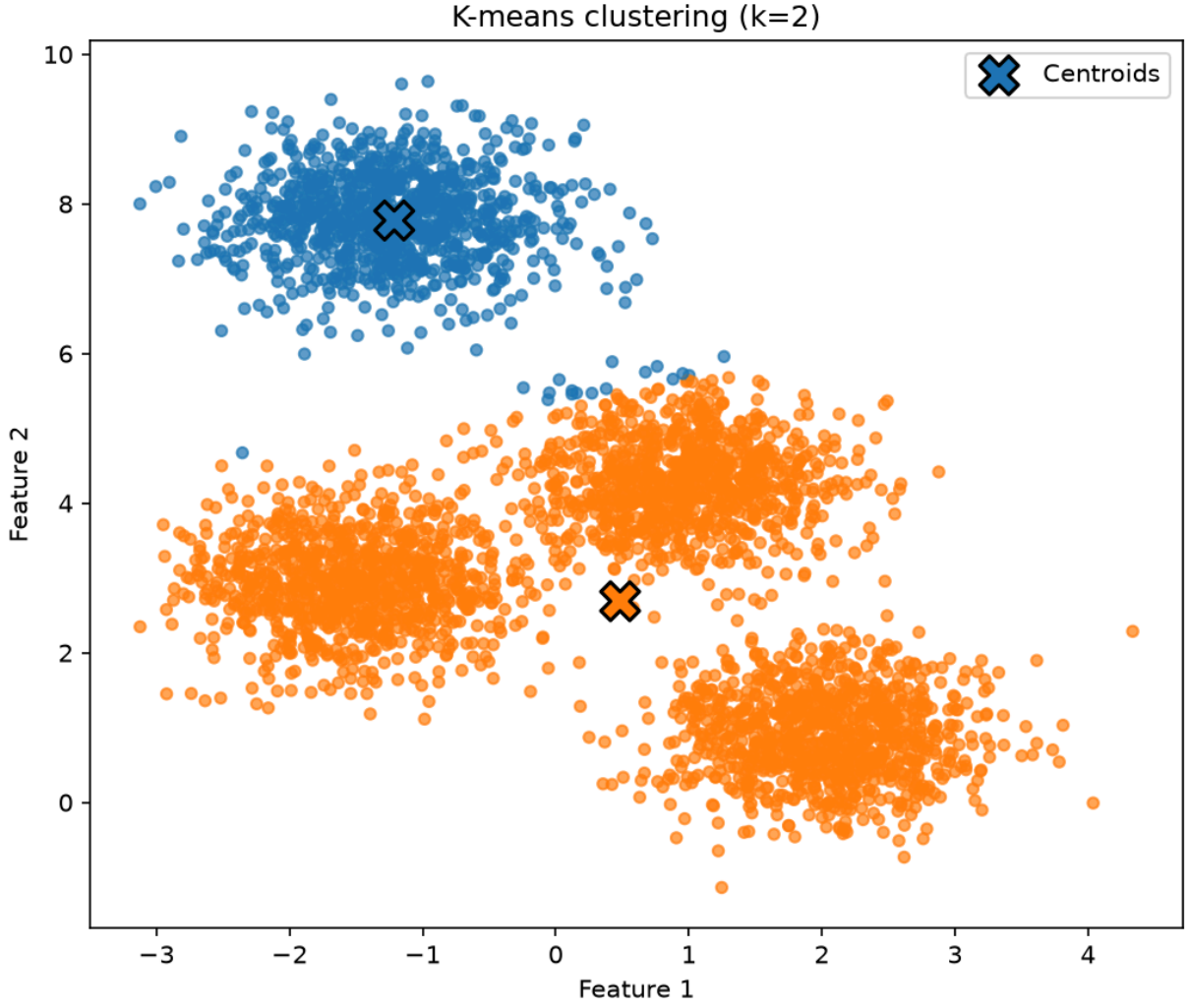
$\begin{pmatrix} 2 \\ 2 \end{pmatrix} x - 3 = 0$

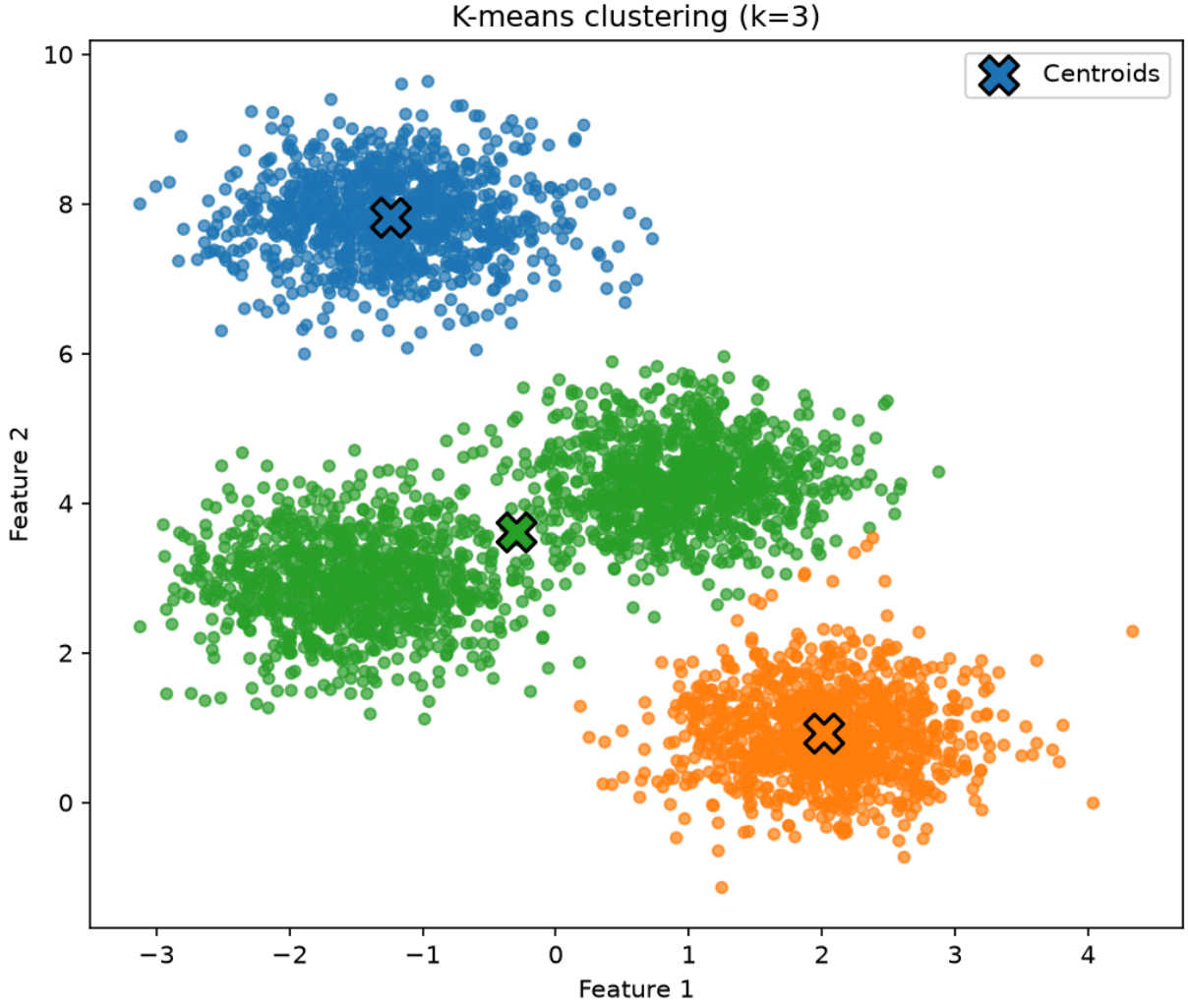
$2x_1 + 2x_2 - 3 = 0$

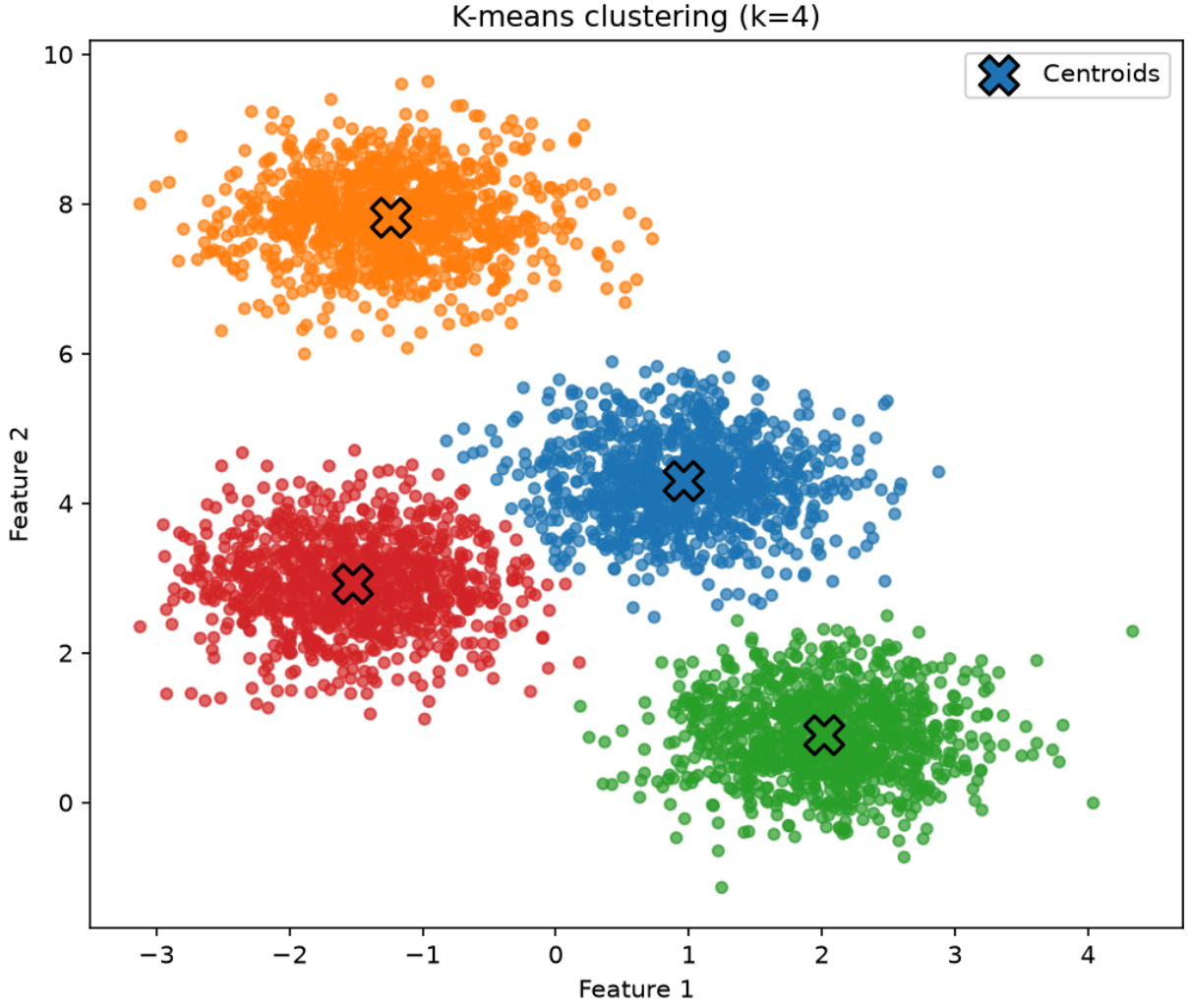
min geometric margin $= \frac{\sqrt{2}}{4}$

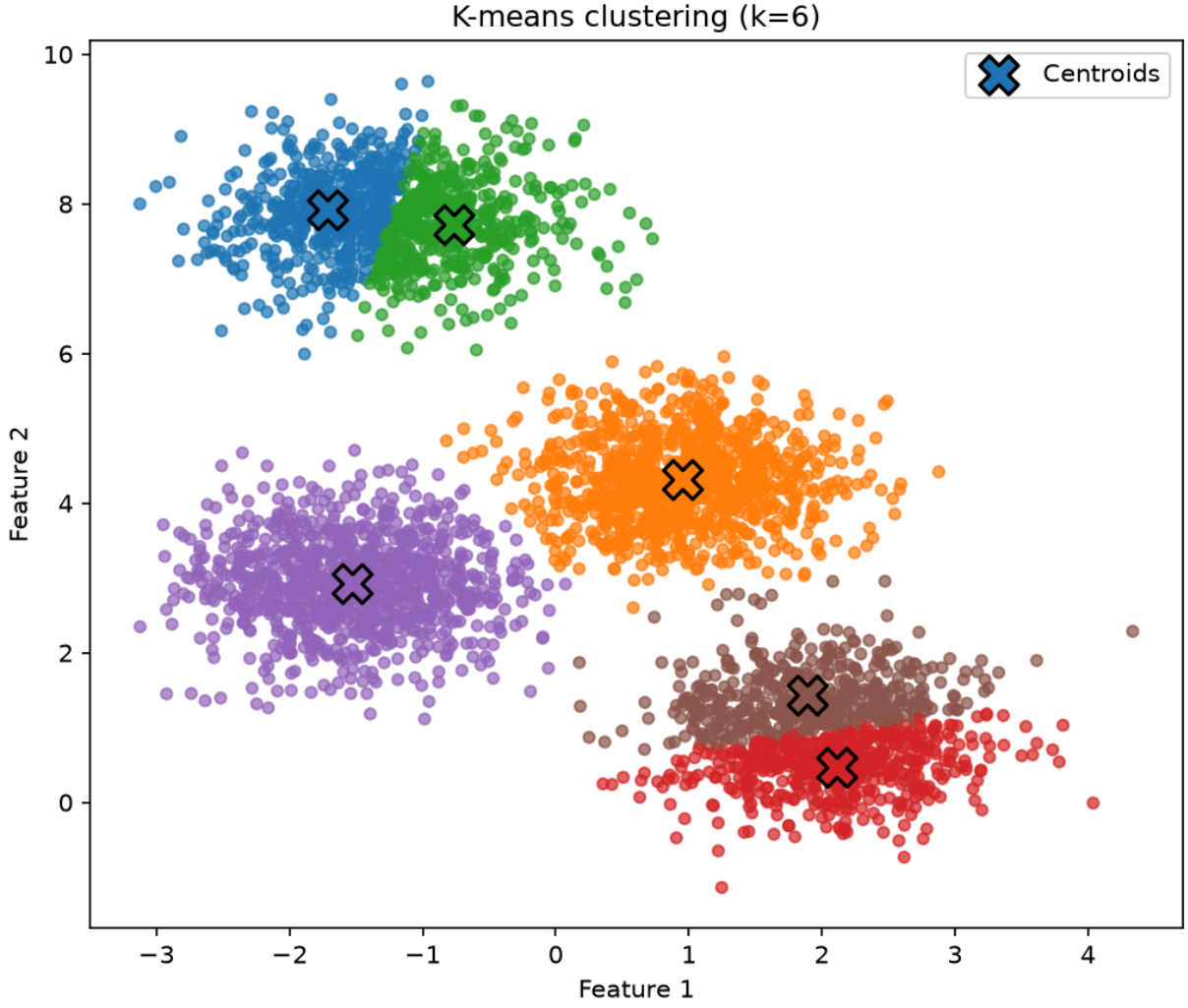

Assignment 3: SVM training data
y = +1
y = -1
support vector
decision boundary
margin
1
2
3
4
5
6
x1
x2
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

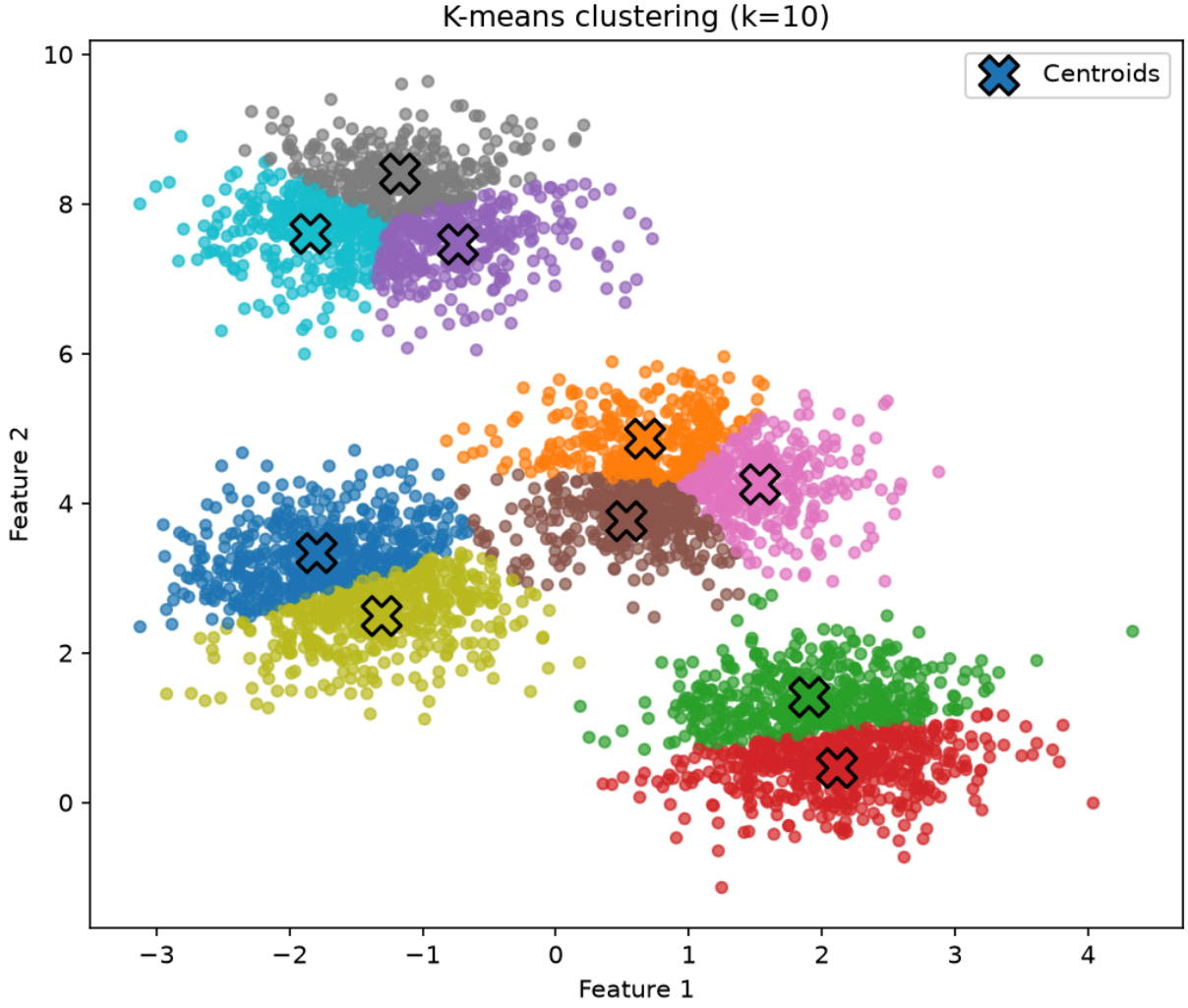
2)

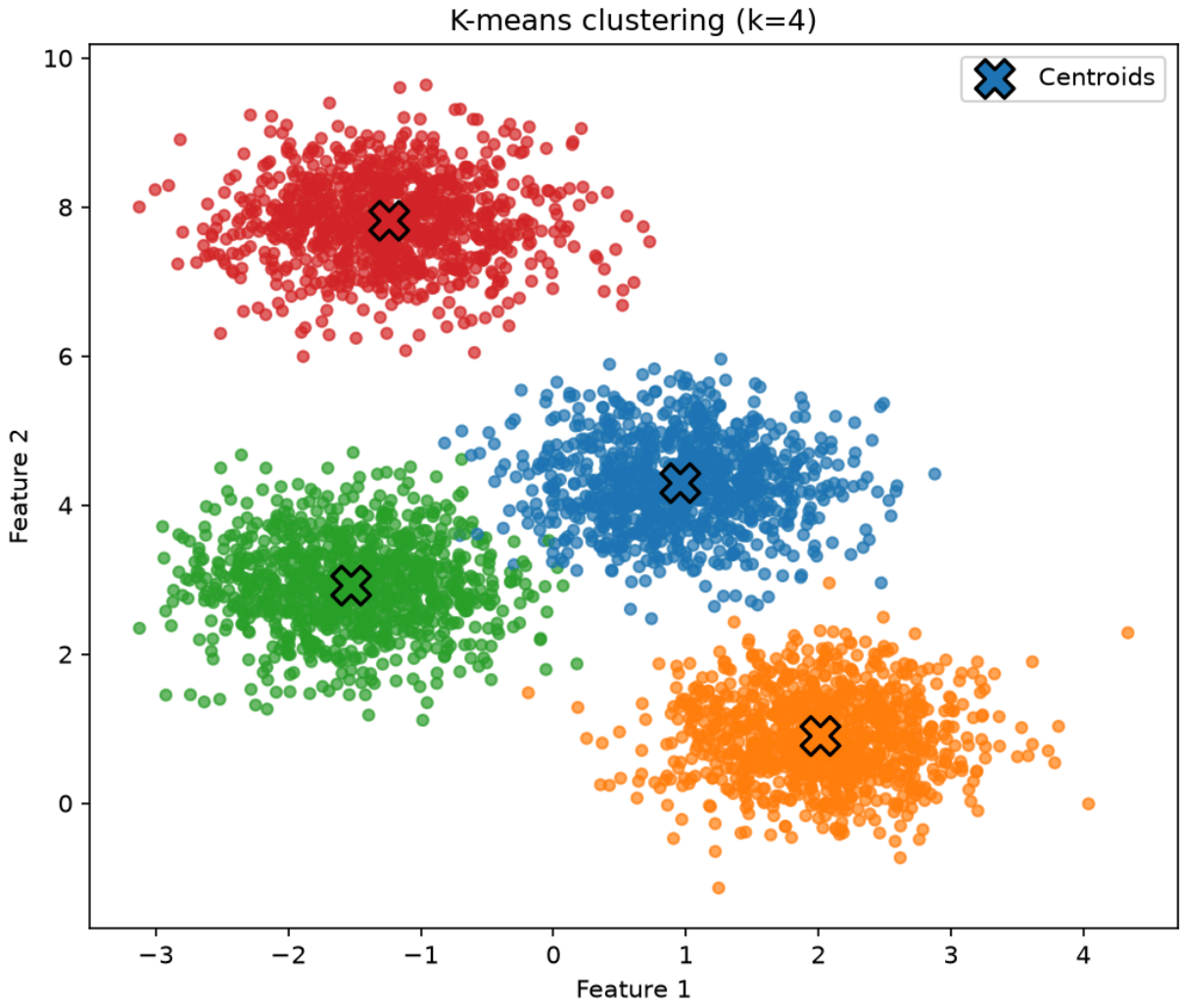

K-means clustering (k=2)
Centroids
Feature 2
Feature 1
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


K-means clustering (k=3)
Centroids
Feature 2
Feature 1
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


K-means clustering (k=4)
Centroids
Feature 2
Feature 1
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


K-means clustering (k=6)
Centroids
Feature 2
Feature 1
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


K-means clustering (k=10)
Centroids
Feature 1
Feature 2
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4



^ Truth

update_assignments: Ties: a point equidistant from several centroids goes to the lowest-index centroid (strict '<' keeps the earlier one).

update_centroids: Empty clusters: a cluster with no assigned points keeps its previous centroid.

To optimize each clustering assignment, I ran 10 restarts for each k value. The K-means algorithm is always trying to reduce the inertia of each cluster

The K-means algorithm is always trying to reduce cluster inertia, so the higher the k is, the lower the inertia is for each cluster, and so is the total inertia

Having a K which is too low groups unrelated clusters into one, while a k which is too high splits clusters into many

k = 2, 3 is underfitting. Clearly distinct clusters are grouped into one.
k = 4 is a perfect match. Each clear cluster has its own centroid.

k > 4 is overfitting. One cluster is split into many.

Increasing k from 2 to 3 and 4 results in very large inertia drops, but increasing beyond results in very low inertia drops, signifying that k=4 is optimal

In this toy scenario, data is specifically made for easy clustering, and we have ground truth to compare to, so it's very easy to tell which K value is optimal.

In a real data set, it is likely that the optimal K value would not be as clear-cut, and different K values might offer different advantages.

3)

KNN increased accuracy by 3.6% on average. Since less than 50% of the data points with each label are flipped during the corruption process, most corrupt points' most neighbors still have the correct label, so running the KNN algorithm helps bring them back to their neighbors' (and likely) their own original label. If the flip rate was closer to 50%, KNN would not help at all. This exercise demonstrates how KNN can be used for anomaly detection